The covar and orig Data-Generating Processes: A Component-Wise

Assessment, with Application to the c_bm <= 0.35 Regime

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1. Executive summary

Paper 01 (`analysis/report/01-dgp-mean-moderation-vs-mvn/`) compares three data-generating processes, of which two are covariance-channel processes: `covar` (the package's Architecture B) and `orig` (a vendored copy of the Hendrickson et al. reference implementation). Appendix B.6 of that paper states that `orig` differs from `covar` on three rows simultaneously. This white paper audits that claim against the code that runs, assesses each difference on its merits, and answers a specific practical question: if the biomarker-response correlation `c_bm` is restricted to 0.35 or below, is there any remaining reason to prefer `covar` over `orig`?

There are five differences, not three, and the appendix describes the wrong version of the most consequential one. The three documented rows are the within-factor correlation form, the cross-factor off-diagonal, and the interaction vector `b`. Two further differences are undocumented: `orig` never assigns the biomarker-response correlation at the first measurement occasion, and it decays the response mean at twice the nominal rate through a `scalefactor` parameter that `covar` lacks. The appendix's description of `b` corresponds to code that is commented out in the vendored file; the active code behaves in the opposite direction.

The active `b` code is the better of the two, contrary to first appearance. The commented-out published version raises the off-drug biomarker-response correlation to full strength as carryover grows, inverting the effect residual exposure should have. The 2024 revision repairs this. Its defects are the `scalefactor` default and the dropped occasion, not the change of gate itself (Section 3.3.1).

Only one of the four covariance-matrix changes is required to fix positive definiteness. A crossed decomposition over all 16 combinations of those four (Section 5) shows that switching compound symmetry to AR(1) accounts for the entire improvement, raising the worst-cell `c_bm` ceiling from 0.34 to 0.48. Applied alone it beats the full four-change `covar` set, which reaches only 0.45. The cross-factor change lowers the ceiling to 0.16 when applied without AR(1), and neither the `b` change nor the occasion-1 change moves it at all.

The answer to the practical question is yes, on five independent grounds. Restricting `c_bm` to 0.35 does not rescue `orig`. The positive-definite (PD) ceiling under `orig` is 0.30 in five of nine design-by-half-life cells and 0.35 in the remaining four, so a study run at 0.35 is non-PD in a substantial part of its own grid. Where it is PD at 0.35 the margin is thin enough to be fragile to any other parameter change. PD failure is repaired silently, so the realized correlation stops matching the nominal one without any warning. The dropped first occasion is a defect at every `c_bm`, including zero. And the `scalefactor` default halves the effective carryover half-life, so the two processes are not comparable at a matched nominal half-life regardless of correlation strength.

The remedy is a corrected `orig`, not a choice between the two. Three of the five 2024 changes are sound and should be kept; the `scalefactor` default and the dropped occasion should be fixed; and the single AR(1) change of Section 5 resolves positive definiteness outright (Section 4.5).

2. Scope, method, and epistemic status

Findings below are labeled `verified` (code was executed and output observed), `inspected` (read in source and confirmed by reading), or `inferred`. Nothing in this document is asserted from memory of the manuscripts.

Sources examined:

- `analysis/scripts/quick-sim/hendrickson-original-comparison/` (`vendored-hendrickson-generateData.R`, `01-hendrickson-orig-driver.R`, `02-pd-sweep.R`, `README.md`)
- `R/generateData.R` (`buildSigma`, the `covar` construction)
- `implementations/original/R/generateData.R`
- `analysis/report/01-dgp-mean-moderation-vs-mvn/report.Rmd`, Appendix sections B.4 through B.7
- The upstream repository `github.com/rchendrickson/pmsimstats`, cloned in full (21 commits, 2020-02-22 to 2024-11-27)

Computations run for this document:

- Reconstruction of the `b` vector under both the active and the commented-out code paths, Hybrid path A, at `t_half` in {0.5, 1.0}.
- A PD sweep over `c_bm` in [0.10, 0.60] by 0.05, crossed with `t_half` in {0, 0.5, 1.0} and the three trial designs, recording the minimum eigenvalue of Sigma under each process, minimized across the design's randomization paths.
- Quantification of the distortion introduced by PD repair.
- A crossed decomposition over all 16 combinations of the four covariance-matrix changes (Section 5).
- Comparison of the carryover mean adjustment at `scalefactor` 1 and 2 against a correct exponential decay from the last on-drug value (Section 3.5).

- Diffs of the vendored files against the upstream initial commit (`42ac030`), the last pre-acceptance commit (`3035581`), and HEAD (`06dac83`).

Provenance correction. The directory README states the vendored files come from `3035581` (dated there as 2026-06-28; the commit is actually 2020-06-27). They do not. Diffing establishes the base as HEAD, `06dac83`, 2024-11-27, from which the vendored copy differs by exactly the one documented `data.table` line. The README's claim that no other lines were changed is therefore true of the file that was vendored and false of the file it names. The `RON THOMAS` annotations are upstream, introduced by RC Hendrickson in commit `8609f12` (2024-05-06, 'Ron Thomas' edits, currated into working code'), not local edits.

All computations used the package's own `extracted_rp` and `extracted_bp` parameter sets, `rho = 0.7`, `c.cf1t = 0.2`, `c.cfct = 0.1`, `N = 35`, `scalefactor = 2`.

3. The five differences

3.1 Difference 1: within-factor correlation form

Status: verified.

`orig` assigns a single constant to every within-factor occasion pair (`vendored-hendrickson-generateData.R:117-126`):

```
ac <- modelparam[[paste("c", c, sep = ".")]]
for (p in 1:(nP-1)) for (p2 in (1+p):nP) {
  correlations[n1, n2] <- ac
  correlations[n2, n1] <- ac
}
```

This is compound symmetry. `covar` uses `rho^|w_i - w_j|`, an AR(1) form on the cumulative week scale.

Arguments for compound symmetry. It is the form in the published reference, so it is the correct target if the goal is reproducing Hendrickson et al. It has one parameter and is exchangeable, which makes it invariant to occasion ordering and therefore robust to irregular visit spacing. It is the standard random-intercept implication, so it is internally consistent with an analysis model that fits only a patient random intercept.

Arguments against. It is clinically implausible for repeated symptom measurement, where adjacent occasions should correlate more strongly than distant ones. More seriously for present purposes, it constrains the feasible parameter space. A constant `rho` across `n` occasions requires `rho > -1/(n-1)` on its own, and once the biomarker-response block is added the joint constraint on `c_bm` tightens rapidly with `n`. Appendix B.7 identifies this as the mechanism behind the narrower PD range, and Section 4 below quantifies it.

Assessment. This difference is the dominant driver of the PD behavior and therefore of the answer to the paper's practical question. It is a legitimate modeling choice in isolation, but it is not a neutral one: it changes what parameter values can be studied at all.

3.2 Difference 2: cross-factor off-diagonal

Status: verified.

`orig` assigns `modelparam$c.cfct` as a constant to every cross-factor pair at distinct occasions (line 140). `covar` assigns `c.cfct * rho^tg`, decaying in the lag `tg`.

Arguments for the constant form. Again, fidelity to the reference. It also imposes no assumption about how quickly the coupling between distinct latent factors decays, which is a quantity for which little empirical guidance exists.

Arguments against. It is internally inconsistent with any decaying within-factor structure, since it asserts that a factor's correlation with a different factor eight occasions away is as strong as with itself at the adjacent occasion. It contributes to the PD constraint in the same direction as Difference 1, since constant off-diagonal blocks raise the matrix's dependence on its smallest eigenvalue.

Assessment. Secondary to Difference 1 in magnitude, and in the same direction. It is not separately identifiable from Difference 1 in any comparison run to date, since the two always vary together.

3.3 Difference 3: the interaction vector b

Status: verified, and the paper's description is incorrect.

Appendix B.6 states that `orig` gates the interaction on whether the response mean is nonzero, giving a two-valued step of `{0, c_bm}`, and that at `t_half = 1.0` in the Hybrid design the vector becomes `c_bm * 1_n`, carrying no on-drug versus off-drug contrast at all. The worked example at B.5 prints `b^orig = (0.4500, 0.4500, 0.4500, 0.4500)`.

That describes the following code, which is present in the vendored file but **commented out**, under the header `## Following commented out in Ron Thomas version:`

```
#if(means[which(n1==labels)]!=0){
#  correlations[n1,'bm']<-modelparam$c.bm
#  correlations['bm',n1]<-modelparam$c.bm
#}
```

The code that actually executes, under the header `## RON THOMAS VERSION:`, is line 153:

```
correlations["bm", n1] <- correlations[n1, "bm"] <-
  ifelse(brtest[p],
```

```
ifelse(brmeans[p] == 0, 0, (mm1 / mm0) * modelparam$c.bm),
modelparam$c.bm)
```

where `mm1` and `mm0` are the carryover-adjusted response means at occasions `p` and `p - 1`, and `brtest` is computed before the carryover adjustment, so it is TRUE exactly at off-drug occasions.

Re-running both code paths over the Hybrid path A design gives:

occasion	on drug	adj mu_BR	b active	b commented-out
OL1	yes	4.281	NA	0.45
OL2	yes	9.223	0.4500	0.45
BD1	yes	9.793	0.4500	0.45
BD2	yes	10.187	0.4500	0.45
BD3	no	2.547	0.1125	0.45
BD4	no	0.159	0.0281	0.45
COd	yes	4.281	0.4500	0.45
COp	no	0.017	0.0018	0.45

(`t_half = 1.0`, `scalefactor = 2`, `c_bm = 0.45`.)

The commented-out column reproduces the appendix exactly. The active column does the opposite: off-drug entries decay toward zero rather than rising to `c_bm`, and the on-drug versus off-drug contrast is preserved rather than lost.

The active decay belongs to the same exponential family as `covar`'s, but runs at twice the rate. The ratio `mm1 / mm0` inherits `(1/2)^(scalefactor * t_sd / t_half)` from the carryover adjustment at line 96, and with `scalefactor = 2` this is an effective half-life of `t_half / 2`, compounding multiplicatively across successive off-drug occasions. For comparison, at `t_half = 1.0` and one week off drug, `covar` gives `0.45 * exp(-ln2 * 1) = 0.225` while `orig` gives `0.1125`.

3.3.1 Which of the two is correct

Status: verified. The natural reading of the paragraphs above is that the commented-out code is the correct target and the active code an unwarranted deviation from it. On assessment that reading is wrong, and it should be stated plainly before the arguments are weighed.

The published gate tests `means[...] != 0`, where `means` is the **carryover-adjusted** response mean. The adjustment runs before the correlation loop, so at any off-drug occasion carrying residual effect, the adjusted mean is nonzero, the gate opens, and the occasion receives the **full** `c_bm`. The consequence is that increasing carryover progressively converts off-drug occasions into full-strength on-drug occasions in the covariance channel, until at `t_half = 1.0` in the Hybrid design the vector is `c_bm * 1_n` and no on-drug versus off-drug contrast survives at all.

That behavior is backwards. Carryover should attenuate the biomarker-response coupling at off-drug occasions, not raise it to on-drug strength. The published DGP therefore has the property that more carryover means a *stronger*, not weaker, off-drug interaction signal, which is not a defensible representation of residual drug exposure.

The active code gates on `brtest`, computed before the carryover adjustment (Difference 5, Section 3.5), so it identifies genuinely off-drug occasions, and then grades the correlation downward rather than holding it at full strength. **The direction is correct and the published version was wrong.** Whatever its implementation defects, the 2024 revision repairs a real error rather than introducing one.

3.3.2 Assessment

Arguments for the active form. It fixes the inversion described above, which is the substantive point. It is graded rather than a step, which is the more plausible representation. And it ties the correlation channel to the same mean trajectory the carryover adjustment already modifies, which is internally consistent.

Arguments against. Its decay rate is set by `scalefactor`, whose default of 2 is itself an error (Section 3.5). It drops occasion 1 (Section 3.4). And it is undocumented: nothing in Paper 01, the vendored file's header, or the directory README describes it, so a reader of the manuscript cannot know which behavior the `orig` arm has.

Assessment. The serious finding here is documentary rather than methodological. Paper 01's Appendix B.5 and B.6 describe the 2020 code, which the `orig` arm does not run, and the appendix's causal story for `orig`, that carryover opens its gate and destroys the identifying contrast, is an accurate account of a defect that the running code has already fixed. Any interpretation in Paper 01 resting on that mechanism requires revision. But the fix should not be to restore the 2020 behavior; it should be to describe the 2024 behavior accurately and correct its two remaining implementation flaws.

3.4 Difference 4: the first occasion is never assigned

Status: verified.

The assignment loop at line 147 is guarded by `if (p > 1)`, and `correlations` is initialized as `diag(length(labels))` at line 106. The biomarker-response correlation at occasion 1 is therefore left at exactly zero under `orig`, regardless of `c_bm` and regardless of whether occasion 1 is on drug. In the table above this appears as the `NA` in the OL1 row. `covar` assigns all eight occasions.

Arguments for. None identified. This does not appear to be a modeling choice.

Arguments against. It removes one of eight occasions from the interaction channel, and it does so at an occasion that is on drug in every path of every design examined, so the loss falls on the most informative part of the series. It attenuates `orig`'s realized interaction relative to its nominal `c_bm` at every parameter setting. It also interacts with Difference 3: the `p > 1` guard exists because line 153 needs `p - 1` to form the ratio, so the defect is a direct consequence of the undocumented modification, not an independent bug.

Assessment. A defect rather than a difference in modeling philosophy. It is `c_bm`-independent and therefore is not addressed by restricting the correlation range.

3.5 Difference 5: the scalefactor on the carryover mean

Status: verified.

Paper 01's Appendix B.6 records the carryover adjustment to the response mean as an 'identical line in all three' processes. It is not. `orig` carries a `scalefactor` multiplier that `covar` does not:

```
## orig
brmeans[p] <- brmeans[p] +
  brmeans[p-1]*(1/2)^(scalefactor * d$tsd[p]/carryover_t1half)

## covar
brmeans[p] <- brmeans[p] +
  brmeans[p-1]*(1/2)^(d$tsd[p]/carryover_t1half)
```

`scalefactor` defaults to 2, so `orig` decays the response mean at twice the nominal rate. This is a difference in the **mean** channel, distinct from the covariance-channel differences above, and it affects any power comparison between the two processes even though it does not enter Sigma and so does not appear in the Section 5 decomposition.

The default is an error, not merely a choice. Comparing the adjustment against a correct exponential decay from the last on-drug value, at `t_half = 1.0` in the Hybrid design:

occasion	t_sd	sf = 1	sf = 2	correct
BD3	1	5.094	2.547	5.094
BD4	2	1.273	0.159	2.547
COp	4	0.268	0.017	0.637

As a ratio to the correct value: at `sf = 1` the first off-drug occasion is exact (1.000x) and later occasions over-decay (0.500x, 0.420x). At `sf = 2` even the first occasion is wrong (0.500x), and the fourth is understated by a factor of 38 (0.026x). A dose one week old at a one-week half-life should retain half its effect; the shipped default gives it a quarter.

The parameter is also undocumented in the strict sense: its Roxygen entry reads `@param scalefactor TODO update when understand what this does?`, so it was shipped with a behavior-changing default while recorded as not understood.

The residual over-decay at `sf = 1` is a separate, pre-existing bug inherited from the published code. The recursion multiplies the previous **already-adjusted** value by `(1/2)^(cumulative t_sd)`, double-counting elapsed time. The correct form uses either the interval rather than cumulative `t_sd`, or cumulative `t_sd` applied to the last on-drug mean.

Arguments for. None identified for the default of 2. The parameter itself is a harmless generalization if defaulted to 1.

Arguments against. It breaks a case the published code got right, it compounds an existing over-decay rather than fixing it, and it silently changes the DGP for every caller that does not override it.

Assessment. This is the change that should be removed. Setting `scalefactor = 1` restores correct behavior at the first off-drug occasion at no cost; fixing the recursion would correct the rest.

4. The c_bm <= 0.35 question

4.1 Why the question arises

Compound symmetry constrains the feasible correlation range more tightly than AR(1), as Appendix B.7 notes. If the binding objection to `orig` were only that it cannot represent large `c_bm`, then restricting attention to `c_bm <= 0.35` would neutralize the objection, and the choice between processes would come down to fidelity to the published reference, which would favor `orig`.

4.2 Positive-definiteness sweep

Status: verified. Minimum eigenvalue of Sigma, minimized across each design's randomization paths, at the parameter settings in Section 2.

Largest `c_bm` on the tested grid at which Sigma remains PD:

design	t_half	orig	covar
CO	0.0	0.30	0.60
CO	0.5	0.30	0.60
CO	1.0	0.30	0.60
Hybrid	0.0	0.30	0.45
Hybrid	0.5	0.35	0.50
Hybrid	1.0	0.35	0.50
OL+BDC	0.0	0.30	0.45
OL+BDC	0.5	0.35	0.45

design	t_half	orig	covar
OL+BDC	1.0	0.35	0.50

Minimum eigenvalue at exactly `c_bm = 0.35`:

design	t_half	orig	covar	orig PD
CO	0.0	-0.3069	8.516	no
CO	0.5	-0.3069	8.516	no
CO	1.0	-0.3069	8.516	no
Hybrid	0.0	-0.3140	2.158	no
Hybrid	0.5	0.0831	2.297	yes
Hybrid	1.0	1.2913	2.340	yes
OL+BDC	0.0	-0.3217	2.156	no
OL+BDC	0.5	0.0869	2.300	yes
OL+BDC	1.0	1.4385	2.344	yes

Three observations follow.

First, `c_bm = 0.35` is **not** a safe ceiling for `orig`. Sigma is non-PD in four of the nine design-by-half-life cells: all three CO cells, and the zero-carryover cells of both other designs. The safe ceiling across the full grid is 0.30, not 0.35.

Second, where `orig` is PD at 0.35, it is marginal. Minimum eigenvalues of 0.083 and 0.087 at `t_half = 0.5` sit three orders of magnitude below `covar`'s at the same cells, and two orders below `orig`'s own values at `c_bm = 0.10`. A margin that thin is not robust to changes in `rho`, `c.cfct`, the number of occasions, or the response parameters, none of which were varied in this sweep.

Third, the CO design is uniformly the binding case under `orig`, and its minimum eigenvalue does not vary with `t_half` at all. This is consistent with CO's off-drug occasions being spaced 2.5 weeks or more apart, so that the `orig` decay, at twice `covar`'s rate, has driven the off-drug `b` entries to effectively zero at every half-life examined.

4.3 Silent repair

Status: verified.

Both implementations call `make.positive.definite(sigma, tol = 1e-3)` when the PD check fails, with no warning, no error, and no record in the returned object:

```r
if (makePositiveDefinite) {
  if (!is.positive.definite(sigma)) {
    sigma <- make.positive.definite(sigma, tol = 1e-3)
```

```
  }
}
```

The consequence is that a non-PD cell does not fail. It silently becomes a different DGP. Measuring the realized biomarker-response correlations after repair, for CO path A at `t_half = 1.0`:

nominal c_bm	min eigenvalue	realized max	repaired
0.20	5.5645	0.2000	no
0.25	5.0361	0.2500	no
0.30	3.6264	0.3000	no
0.35	0.8645	0.3500	no
0.40	-2.8964	0.3955	yes
0.45	-7.3385	0.4370	yes

The distortion is modest in magnitude, roughly one to three points, but it is undisclosed and it is not constant across the grid, so it introduces a systematic and unreported difference between nominal and realized effect size precisely in the cells where `orig` is under the most strain. Power reported against a nominal `c_bm` in a repaired cell is not power at that `c_bm`.

4.4 Answer

Restricting `c_bm` to 0.35 or below does not remove the case for `covar`. Five reasons remain, in descending order of force.

1. **The restriction is not tight enough.** At exactly 0.35, `orig` is non-PD in four of nine cells, including every CO cell. A restriction that actually cleared the PD objection would be `c_bm <= 0.30`, and even then only if no other parameter moves.
2. **The `scalefactor` default is `c_bm`-independent and wrong.** `orig` decays the response mean at twice the nominal rate at every correlation strength (Section 3.5), understating residual exposure by a factor of two at the first off-drug occasion and by up to 38 at the fourth. The two processes are therefore not matched on carryover at a matched nominal `t_half`, so any comparison between them confounds carryover severity with everything else.
3. **The dropped first occasion is `c_bm`-independent.** It removes an on-drug occasion from the interaction channel at every parameter setting, including the null.
4. **Silent repair breaks the correspondence between nominal and realized effect size** in exactly the region where a restricted study would be operating closest to the boundary.

5. **The fidelity argument does not survive inspection, but not in the way it first appears.** The strongest reason to prefer `orig` at any `c_bm` is that it reproduces the published reference. The vendored file does not: it is upstream HEAD (`06dac83`, 2024-11-27), four years after the paper, not the pre-acceptance commit the directory README names. So `orig` as run is not Hendrickson et al. as published.

   The remedy is not to restore the published version. Section 3.3.1 shows the 2020 interaction gate is defective: it raises the off-drug biomarker-response correlation to full strength as carryover grows, inverting the effect residual exposure should have. Pinning to the pre-acceptance commit would buy documentary fidelity at the cost of reinstating a scientifically wrong DGP.

Section 5 qualifies the first of these five reasons. A crossed decomposition shows the PD objection is narrower than it appears: it is an objection to compound symmetry alone, and one change fixes it. The other four reasons are not PD arguments and are unaffected.

The fifth point deserves the most emphasis, because it dissolves the trade-off rather than resolving it. One would ordinarily accept a narrower feasible parameter range as the price of matching a published method. Here neither branch is attractive: the current arm does not match the published method, and the published method is not worth matching. The way out is a corrected process rather than a choice between the two existing ones, which is the subject of Section 4.5.

4.5 What would change the answer

Rather than choosing between the two existing processes, the better course is a corrected `orig`. Assessing the five 2024 changes individually (Sections 3.3 to 3.5) shows they are not of a piece: three are sound and two are defective.

Keep. The `brtest` / `rawbrmeans` capture (Difference 5's infrastructure), which correctly separates pre-carryover state from post-carryover mean. The interaction gate replacement, which repairs the inversion in the published code (Section 3.3.1). The `verbose` diagnostic blocks, which are inert with respect to returned values and are the mechanism by which the `scalefactor` problem would have been caught.

Fix. Set `scalefactor = 1`, or remove the parameter (Section 3.5). Remove the `p > 1` guard so occasion 1 is assigned (Section 3.4), which requires handling `p = 1` explicitly since the ratio has no predecessor there.

Fix separately. The carryover recursion double-counts elapsed time by applying `(1/2)^(cumulative t_sd)` to an already-adjusted value. This is inherited from the published code and is present in `covar` as well, so correcting it changes both processes.

A process so corrected would be better than the published original, better than current HEAD, and would isolate the correlation structure as the substantive remaining difference from `covar`. Combined with the single AR(1) change of Section 5, it would also resolve the PD objection outright.

Two further conditions would independently strengthen any comparison:

- Restricting the study to `c_bm <= 0.30` and to designs with `t_half > 0`, under which the PD objection does not arise even without the AR(1) change.
- Running the correlation structure and the interaction gate as crossed factors, as Paper 01's own Appendix B.6 suggests. Section 5 does this for positive definiteness; the power half remains undone.

5. Decomposition: the minimum set of changes that fixes PD

5.1 Motivation and method

The stated reason for the `covar` changes was the non-positive-definite correlation matrix under `orig`. That justification implies a counterfactual which has not previously been tested: how many of the four changes are actually required to fix PD, and does each contribute in the direction assumed?

Status: verified. A single Sigma constructor was written with one flag per candidate change, reproducing `orig` when all four flags are off and `covar` when all four are on. The flags are:

- `ar1`: AR(1) within-factor (`rho^|w_i - w_j|`) versus compound symmetry (constant `rho`). Difference 1.
- `cfd`: decaying cross-factor off-diagonal (`c.cfct * rho^|w_i-w_j|`) versus constant `c.cfct`. Difference 2.
- `bexp`: exponential `b` (`c_bm * exp(-lambda * t_sd)`) versus the ratio-graded `orig` form. Difference 3.
- `occ1`: assign the biomarker-response correlation at occasion 1 versus skipping it. Difference 4.

All 16 combinations were crossed with three designs and three half-lives. For each, `c_bm` was swept from 0.05 to 0.80 in steps of 0.01, and the largest value at which Sigma remained PD across every randomization path was recorded. The reported ceiling for a configuration is its worst cell over all nine design-by-half-life combinations.

5.2 Effect of each change in isolation

Starting from `orig` (all four flags off, worst-cell ceiling 0.34) and turning on exactly one flag:

change enabled	worst-cell ceiling	effect
none (`orig`)	0.34	baseline

change enabled	worst-cell ceiling	effect
`ar1` only	**0.48**	**+0.14**
`cfd` only	0.16	-0.18
`bexp` only	0.34	none
`occ1` only	0.34	none

The result is unambiguous. **The AR(1) change alone accounts for the entire PD improvement.** The other three contribute nothing positive, and one of them is actively harmful.

5.3 The minimum set is a single change

`ar1` is both necessary and sufficient. Every configuration with `ar1 = TRUE` reaches a worst-cell ceiling of 0.43 or above; every configuration with `ar1 = FALSE` is capped at 0.34 or below, regardless of what the other three flags are set to. Ranked across all 16 configurations, the eight `ar1 = TRUE` configurations occupy the top eight positions without exception.

Per design and half-life:

design	t_half	orig	AR(1) only	full covar
CO	0.0	0.34	0.59	0.61
CO	0.5	0.34	0.59	0.61
CO	1.0	0.34	0.59	0.61
Hybrid	0.0	0.34	0.49	0.45
Hybrid	0.5	0.35	0.51	0.50
Hybrid	1.0	0.36	0.54	0.54
OL+BDC	0.0	0.34	0.48	0.45
OL+BDC	0.5	0.35	0.49	0.49
OL+BDC	1.0	0.36	0.52	0.53

Changing only the within-factor correlation form, and leaving the other three `orig` behaviors untouched, yields a higher worst-cell ceiling (0.48) than the full set of four changes (0.45). The minimal fix is not merely sufficient; it is strictly better than `covar` on the criterion that motivated `covar` in the first place. AR(1)-only matches or exceeds full `covar` in seven of the nine cells, and trails it by at most 0.01 in the remaining two.

5.4 Why the cross-factor change is harmful

The `cfd` change, applied without `ar1`, lowers the ceiling from 0.34 to 0.16, the worst configuration tested. The mechanism is structural incoherence. Under compound symmetry a factor's correlation with itself is a constant `rho` at every lag. Introducing a decaying cross-factor term makes a factor's correlation with a *different* factor fall away with lag while its correlation with *itself* does not. At long lags the matrix asserts strong within-factor coupling alongside near-zero cross-factor coupling, and the resulting block structure drives the smallest eigenvalue negative sooner than either uniform choice does.

Under `ar1` the same change is close to neutral, contributing at most 0.02 in either direction, because within-factor and cross-factor terms then decay together and the matrix stays internally consistent.

This is worth stating plainly: **Difference 2 was not a PD fix. In isolation it is a PD regression, and its only defensible justification is internal consistency with Difference 1.**

5.5 Why the b and occasion-1 changes are irrelevant to PD

Neither `bexp` nor `occ1` moved the ceiling by a single grid step in any configuration. This is expected on inspection. The biomarker-response entries occupy one row and one column of a matrix with `2 + 3n` dimensions, and their magnitudes are bounded by `c_bm` itself. The binding constraint comes from the `3n x 3n` within- and cross-factor blocks, not from the biomarker margin.

The practical consequence is that Differences 3 and 4 cannot be defended on PD grounds at all. Difference 3 is an undocumented change of decay rate (Section 3.3) and Difference 4 is a defect (Section 3.4). Whatever their merits, the non-positive-definiteness of the `orig` matrix is not among them.

5.6 Implications

1. **The minimum set of changes that fixes the PD problem is one change: compound symmetry to AR(1).** Adopting it alone recovers more feasible parameter range than the full four-change set.
2. **Three of the four changes in `covar` are not PD fixes.** They may be justified on other grounds, but the PD rationale does not extend to them and should not be offered for them.
3. **A minimal-change variant is worth constructing.** An `orig + AR(1)` process would isolate the correlation structure as the sole difference from the published reference, which is exactly the crossed-factor decomposition Paper 01's Appendix B.6 says would be required, and it is a one-line change to the vendored file.
4. **This changes the practical answer of Section 4.4 in one respect.** The PD objection to `orig` is narrower than it appeared: it is an objection to compound symmetry specifically, not to the `orig` process as a whole. The other four objections in Section 4.4 stand unchanged, since none of them is a PD argument.

6. Recommendations

1. **Do not adopt `orig` as a comparison arm on the strength of a restricted `c_bm` range.** The restriction addresses one of four differences and does not fully address even that one.

2. **Correct Paper 01, Appendix B.5 to B.7.** The `b` row, the worked example, and the third interpretive paragraph after the B.6 table describe code that does not run. This is independent of any decision about future simulation work and should be treated as a correctness fix.

3. **Correct the directory README's provenance paragraph.** The vendored base is upstream HEAD `06dac83` (2024-11-27), not `3035581`, and the date given for that commit (2026-06-28) is wrong in both year and day (it is 2020-06-27). The 'no other lines were changed' claim is accurate once the correct base is named. The `RON THOMAS` annotations are upstream, from `8609f12`, and should not be described as local modifications.

3b. **Do not revert to the published version to resolve this.** Section 3.3.1 shows the 2020 interaction gate is defective. Pinning to `3035581` would buy documentary fidelity at the cost of a scientifically wrong DGP. Correct the description and the two remaining implementation flaws instead (Section 4.5).

4. **Make PD repair loud.** Both implementations should record the repair in the returned object and warn. A silent change to the DGP is not acceptable in a simulation study whose outputs are reported against nominal parameter values.

5. **Fix the `p > 1` guard** if `orig` is retained in any form, or document it as a known property of the arm if the intent is to reproduce whatever the vendored file does.

6. **Construct an `orig + AR(1)` variant.** Section 5 shows this is a one-line change that fixes PD more effectively than the full `covar` set. It isolates the correlation structure as the sole difference from the reference implementation and would let Paper 01 report a decomposition rather than a confounded bound.

7. **Stop offering the PD rationale for Differences 2, 3, and 4.** Section 5 shows none of them improves positive definiteness, and Difference 2 degrades it when applied on its own. If those changes are retained they need a different justification.

8. **If a full decomposition of the power results is wanted, run the crossed design.** Correlation structure (CS versus AR(1)) crossed with interaction gate (step versus graded) at `c_bm <= 0.30`, three designs, three half-lives. This is four DGP variants rather than two and would separate the effects the present comparison confounds. The PD half of that decomposition is already done (Section 5); the power half is not.

7. Limitations

The PD sweep used a single parameter set (`rho = 0.7`, `c.cf1t = 0.2`, `c.cfct = 0.1`, `N = 35`, eight occasions, the package's `extracted_rp` and `extracted_bp`). The PD boundary depends on all of these, and the boundaries reported in Section 4.2 should not be read as general. In particular, larger `rho` will tighten the `orig` boundary further.

The Section 5 decomposition is a statement about positive definiteness only. It establishes which changes widen the feasible `c_bm` range, not which produce better statistical behavior within that range. An `orig + AR(1)` process has a higher PD ceiling than `covar` but was not evaluated for power, Type I error, or estimator bias, and nothing here implies it would perform better on those.

The Section 5 constructor reproduces both processes from the source read in Section 2 rather than calling them. It was checked against the `orig` and `covar` ceilings obtained in Section 4.2 from the real implementations, which agree to within the coarser grid step used there, but it is a reimplementation and may diverge in behavior not exercised by this sweep.

No power simulation was run for this document. The consequences of Differences 3 and 4 for realized power under `orig` are argued from the structure of the covariance matrix, not measured. Quantifying them would require running the `orig` arm with and without the `p > 1` guard, and with the commented-out gate restored, which is the natural follow-up.

The upstream provenance question is now settled by direct comparison (Section 2), but one part of it is not. Commit `8609f12` is authored and committed by RC Hendrickson under the message 'Ron Thomas' edits, currated into working code'. The history cannot show which parts of that commit originated with Ron Thomas and which arose in the curation, so responsibility for the `scalefactor` default and the `p > 1` guard cannot be assigned from the record. This matters only for deciding with whom to raise them.

The upstream code has not been executable on a current stack for some time: the pre-2021 `data.table` indexing idiom is present in all 21 commits including HEAD, so `generateData()` fails outright without the vendored patch. The `orig` arm therefore reproduces upstream source but not upstream behavior, since upstream does not run unmodified.

`implementations/original/R/generateData.R`, the package's other copy of the reference implementation, was inspected and found to differ from the vendored copy: it gates on `d[p]$onDrug` and uses `c.cfct * rho^tg` for the cross-factor off-diagonal. It is therefore a third variant, distinct from both `covar` and the vendored `orig`, and its relationship to the other two was not pursued here.

8. Provenance

Audit conducted 2026-09-08 against the working tree at commit `3ce1162`. Computations executed on the host R session (R 4.6.1), outside the project container. Scratch scripts were not retained in the repository.